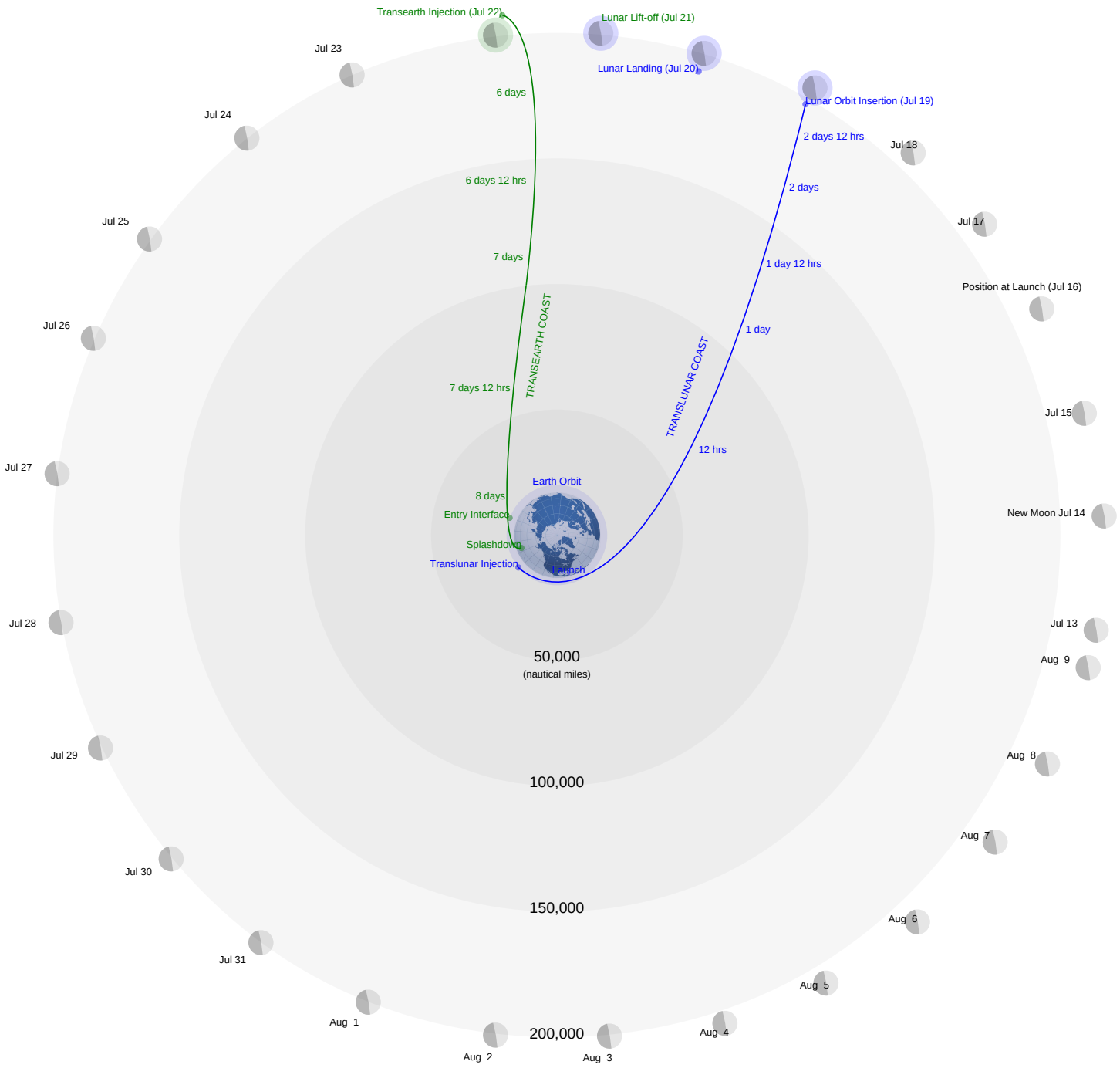
APOLLO 11 TRANSLUNAR / TRANSEARTH TRAJECTORY PLOTTING CHART



This chart displays a polar view of the lunar mission profile for July 16, 1969. Translunar injection occurs over the Pacific Ocean in the vicinity of the Solomon Islands during the second revolution. The profile is typical of the Apollo 11 mission planned for July and August 1969. Each month contains three possible launch days with each day targeted to a specific Apollo landing site.

Launch Dates		Site No.	Site Coordinates	
July 16	Aug 14	2	0°42.8'N	23°42.5'E
July 18	Aug 16	3	0°21.2'N	1°18.0'W
July 21	Aug 20	5	1°40.7'N	41°54.0'W

Periodically, during the translunar coast phase, mid-course correction may be performed to maintain a circumlunar trajectory and ensure that the spacecraft will remain within a 60 nautical mile orbit above the moon. This orbit will be maintained by the Command Service Module (CSM) throughout the Lunar Module (LM) landing and lift-off phases. Midcourse corrections, if required, during the transearth coast phase will enable the spacecraft to approach the atmosphere at a proper velocity and angle for a safe re-entry. Elapsed time to flight from the translunar injection point is indicated during the translunar and transearth coast phases.

Event	Time (EDT)
Launch	July 16 9:32
Earth Orbit	July 16 9:43
Translunar Injection	July 16 12:16
Translunar coast	July 16-19
Lunar Orbit Insertion	July 19 13:21
Lunar Landing	July 20 16:17
Lunar Lift-off	July 21 13:54
Transearth Injection	July 22 00:55
Transearth coast	July 22-24
Entry Interface	July 24 12:39
Splashdown	July 24 12:50

Approximate flight time for a July 16, 1969 launch date is as follows:

	hrs	min
Launch to Translunar Injection		
Translunar Injection to Lunar Orbit Insertion	2	44
Lunar Orbit Insertion to Transearth Injection	73	16
Transearth Injection to Entry Interface	59	24
Entry Interface to Pacific Ocean Touchdown	59	37
	0	14
Total Flight Time	195	15